BUNNY SKY

Game Design Document · Game Development Fundamentals · RK2 · Kadirbek

> **Pitch:** A blaster-wielding bunny hops across floating sky islands, collects coins and fights the monsters that took over the sky, ending each world with a boss.

1. Genre · Platform

- **Genre:** 3D action platformer (third-person)
- **Platform:** Android (APK), phone and tablet
- **Orientation:** landscape
- **Session:** 2–4 minutes per level

2. 2D or 3D

- **3D**, camera behind and above the hero
- Hero: a real **rigged character** (blue bunny with a blaster) with Idle, Run, Jump, Wave animations
- Movement relative to the camera; the hero turns to face the direction

3. Core loop

RUN & JUMP → COLLECT coins → FIGHT enemies → REACH the flag → next island ↺

The flag opens only when all coins of the level are collected, so the player explores the whole level before moving on.

4. Controls

On-screen control	Where	What it does	Keyboard
Joystick	bottom-left	move the hero relative to the camera	WASD / arrows
JUMP button	bottom-right	jump (only from the ground)	Space
FIRE button	bottom-right	shoot a blaster bullet (cooldown 0.4 s)	X
Pause button (II)	top-right	open the pause screen, time stops	—
Menu / Pause buttons	center	Play, Quit · Resume, Restart, Menu	—

5. Content

Type	Now (midterm)	Final (week 15)
Levels	2 levels + 1 boss arena	4 levels + 1 boss arena
Enemies	Crab (patrol), Bee (flying patrol), Boss Horned King (fan of fireballs, 2 phases)	+ Skull Chaser (follows the player when close — enemy AI)
Hazards	spikes, spinning saw, falling into the sky	+ moving saws
Items	coins (score), hearts (+1 HP)	+ star per level (saved best result)
Mechanics	3 HP, blinking invincibility, lift platform	+ saved progress, level select

6. Art and sound

- **Style:** low-poly cartoon, bright sky islands, soft shadows — one world from menu to win screen.
- **Assets:** Ultimate Platformer Pack by Quaternius (CC0) for the hero, enemies, boss, items and islands; UI icons rendered from the same models, so the HUD matches the game.
- **Sound:** 8-bit effects (jump, coin, shoot, hurt, heal, boss hit, win) and two music loops (levels, boss), music quieter than effects.

7. MVP — week-15 build

- Main menu, HUD, pause screen (safe area, all screens)
- 4 levels + boss + win screen, death restarts the level

8. Cut list — will NOT make

- Multiplayer / online leaderboard
- Shop, skins, in-app purchases
- Open world, story cutscenes

- 3 normal enemy types incl. one with chase AI
- Saving: unlocked levels, best coins, volume
- Sound settings, particle effects for coins and hits
- Android APK, 30+ FPS
- Second boss and more than 4 levels
- iOS build (no Mac)

9. Plan · weeks 9–13

Week	Ready at the end of the week
9	GDD, GitHub repository, Main Menu / HUD / Pause with safe area, tested on 3 devices
10	Enemy AI: Skull Chaser (patrol → chase when the player is close → return)
11	Saving: unlocked levels, best coins per level, settings; level select in the menu
12	Sound settings (music / effects volume), particle effects for coins, hits and boss
13	Levels 3 and 4, balancing, bug fixes, Android build

10. Roles

Solo until week 14 — **Kadirbek:** game design, programming, level design, UI, testing on the phone, Git. From week 14, if a team forms: one person on code, one on levels and art, one on UI, sound and testing.